Area and Volumes of Revolution

MATH 182 — Class Notes

October 16, 2026 | 30-minute discussion

Objectives. Sketch a bounded region, choose a slice direction, identify lengths and radii, and set up and evaluate area and volume integrals.

Class sequence: area (6 minutes), disks and washers (9 minutes), shells (9 minutes), practice and review (6 minutes).

1. Start with a slice

An integral adds small pieces. For area, each piece is a thin rectangle. For a volume of revolution, rotate a representative slice and determine the resulting shape.

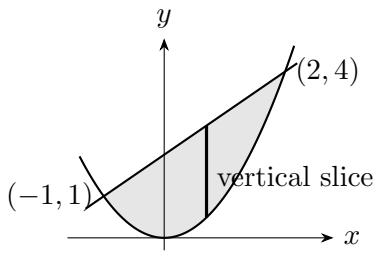
$$\boxed{A = \int_a^b (\text{top} - \text{bottom})\, dx}, \qquad \boxed{A = \int_c^d (\text{right} - \text{left})\, dy}.$$

Setup routine: sketch and shade; find intersections; draw a slice; label its lengths; write the integral with matching limits; evaluate when requested.

Example 1. Area between a line and a parabola

Find the area enclosed by $y = x + 2$ and $y = x^2$.

Intersections: $x^2 = x + 2$ gives $(x - 2)(x + 1) = 0$, so $x = -1, 2$. The line lies above the parabola on this interval.



The slice has height $(x + 2) - x^2$ and thickness dx. Therefore

$$A = \int_{-1}^{2} (x + 2 - x^2)\, dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \boxed{\frac{9}{2} \text{ square units}}.$$

Check your understanding. Why must we subtract the parabola from the line? Because a slice length is top minus bottom and must be nonnegative.

2. Disks and washers: perpendicular slices

A slice perpendicular to the rotation axis sweeps out a disk or washer. The outer radius R and inner radius r are distances from the axis.

$$\boxed{V = \pi \int (R^2 - r^2)\, dt}, \qquad \text{disk: } r = 0, \quad dt = dx \text{ or } dy.$$

Axis guide. About a horizontal axis, use vertical slices (dx). About a vertical axis, use horizontal slices (dy).

Example 2. Disk method

Rotate the region under $y = \sqrt{4 - x}$, above the x-axis, for $0 \le x \le 4$, about the x-axis.

A vertical slice touches the rotation axis, so it creates a disk. Its radius is $R(x) = \sqrt{4 - x}$.

$$\begin{aligned} V &= \pi \int_0^4 \left(\sqrt{4 - x}\right)^2 dx = \pi \int_0^4 (4 - x)\, dx \\ &= \pi \left[4x - \frac{x^2}{2}\right]_0^4 = \boxed{8\pi \text{ cubic units}}. \end{aligned}$$

Example 3. Washer method

Rotate the region between $y = x + 1$ and $y = 1$, for $0 \le x \le 2$, about the x-axis.

The slice does not reach the axis, so there is a hole. The outer radius is $R = x + 1$; the inner radius is $r = 1$.

$$\begin{aligned} V &= \pi \int_0^2 \left[(x + 1)^2 - 1^2\right] dx \\ &= \pi \int_0^2 (x^2 + 2x)\, dx = \pi \left[\frac{x^3}{3} + x^2\right]_0^2 = \boxed{\frac{20\pi}{3} \text{ cubic units}}. \end{aligned}$$

Essential distinction: subtract the areas of two disks:

$$\pi R^2 - \pi r^2 = \pi(R^2 - r^2), \qquad \text{not } \pi(R - r)^2.$$

Quick extension: a shifted axis

If Example 3 is rotated about $y = -2$, measure each distance from that line:

$$R = (x + 1) - (-2) = x + 3, \qquad r = 1 - (-2) = 3.$$

The setup becomes

$$\boxed{V = \pi \int_0^2 \left[(x + 3)^2 - 3^2\right] dx}.$$

Key question: Which boundary is farther from the actual axis? That boundary determines the outer radius.

3. Cylindrical shells: parallel slices

A slice parallel to the rotation axis sweeps out a thin cylindrical shell. Unrolling it gives a rectangle with length equal to its circumference.

$$\boxed{dV = 2\pi(\text{radius})(\text{height})(\text{thickness})}.$$

The shell height is the length of the original slice. For a horizontal slice, this is a horizontal length.

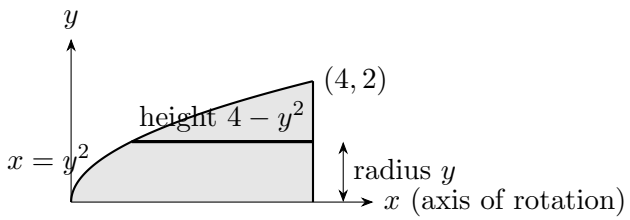
Example 4. Horizontal shells about the x-axis

Rotate the first-quadrant region bounded by $x = y^2$, $x = 4$, and $y = 0$ about the x-axis. Use shells.

Bounds: $y^2 = 4$ gives the upper height $y = 2$. Thus $0 \le y \le 2$.

One horizontal slice:

$$\text{radius} = y, \qquad \text{height} = \text{right} - \text{left} = 4 - y^2, \qquad \text{thickness} = dy.$$



Therefore

$$\begin{aligned} V &= 2\pi \int_0^2 y(4 - y^2)\, dy \\ &= 2\pi \left[2y^2 - \frac{y^4}{4}\right]_0^2 = 2\pi(8 - 4) = \boxed{8\pi \text{ cubic units}}. \end{aligned}$$

Independent check: vertical disks have radius $\sqrt{x}$ for $0 \le x \le 4$, so $V = \pi \int_0^4 x\, dx = 8\pi$.

Choose the direction before writing a formula

Axis	Vertical slices (dx)	Horizontal slices (dy)
Horizontal	Disks/washers	Shells
Vertical	Shells	Disks/washers

Vertical-shell model: about the y-axis, a region with top $f(x)$ and bottom $g(x)$, contained in $x \ge 0$, has

$$V = 2\pi \int_a^b x[f(x) - g(x)]\, dx.$$

4. Practice and final checks

Practice A — washer setup. Rotate the region between $y = 3$ and $y = x$, for $0 \le x \le 3$, about the x-axis. Label R and r, then set up the volume integral.

Practice B — shell setup. Rotate the region under $y = 4 - x$ and above $y = 0$, for $0 \le x \le 4$, about the y-axis. Label the shell radius and height, then set up the volume integral.

Answer checks

A: $R = 3$, $r = x$, and

$$V = \pi \int_0^3 (9 - x^2)\, dx = \boxed{18\pi}.$$

B: radius x, height $4 - x$, and

$$V = 2\pi \int_0^4 x(4 - x)\, dx = \boxed{\frac{64\pi}{3}}.$$

Before submitting a solution

- Does the sketch show the bounded region and the actual rotation axis?
- Are the slice lengths and radii nonnegative?
- Do the limits match the integration variable?
- For washers, did you use $R^2 - r^2$? For shells, did you use $2\pi rh$?
- Does a boundary change inside the interval? If so, split the integral where its formula changes, or choose another slice direction.
- Have you used the requested method and kept the answer exact?

Exit question

For rotation about the y-axis, explain why a vertical slice produces a shell and a horizontal slice produces a washer.

Explanation: the vertical slice is parallel to the axis and sweeps around it at a fixed radius, forming a shell. The horizontal slice is perpendicular to the axis and sweeps out a circular cross-section, forming a disk or washer.